COMPUTER ENGINEERING

QUEZON CITY

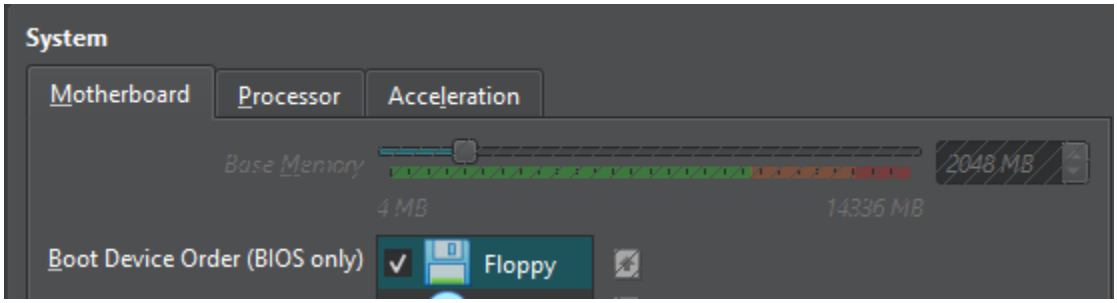
COURSE CODE/TITLE:	CPE 212 | Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	07-23-26	**DATE SUBMITTED:**	07-23-26
NAME:	Masongsong, John Marvin B.	**INSTRUCTOR:**	Engr. Jimlord Quejado

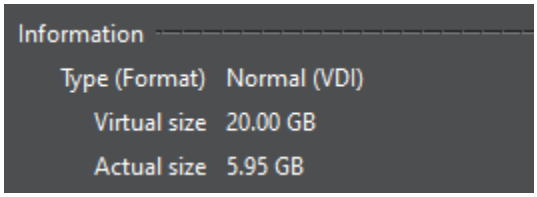
ACTIVITY NO. 3 **Install SSH server on CentOS or RHEL 8**

1. PROCEDURE OUTPUT

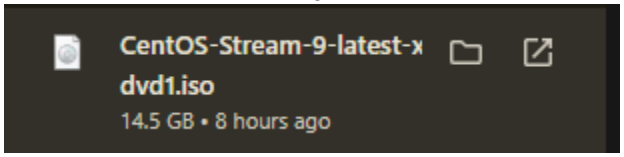
Task 1: Download the CentOS or RHEL-8 image (Create screenshots of the following)

1. Download the image of the CentOS here: http://mirror.rise.ph/centos/7.9.2009/isos/x86_64/
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.

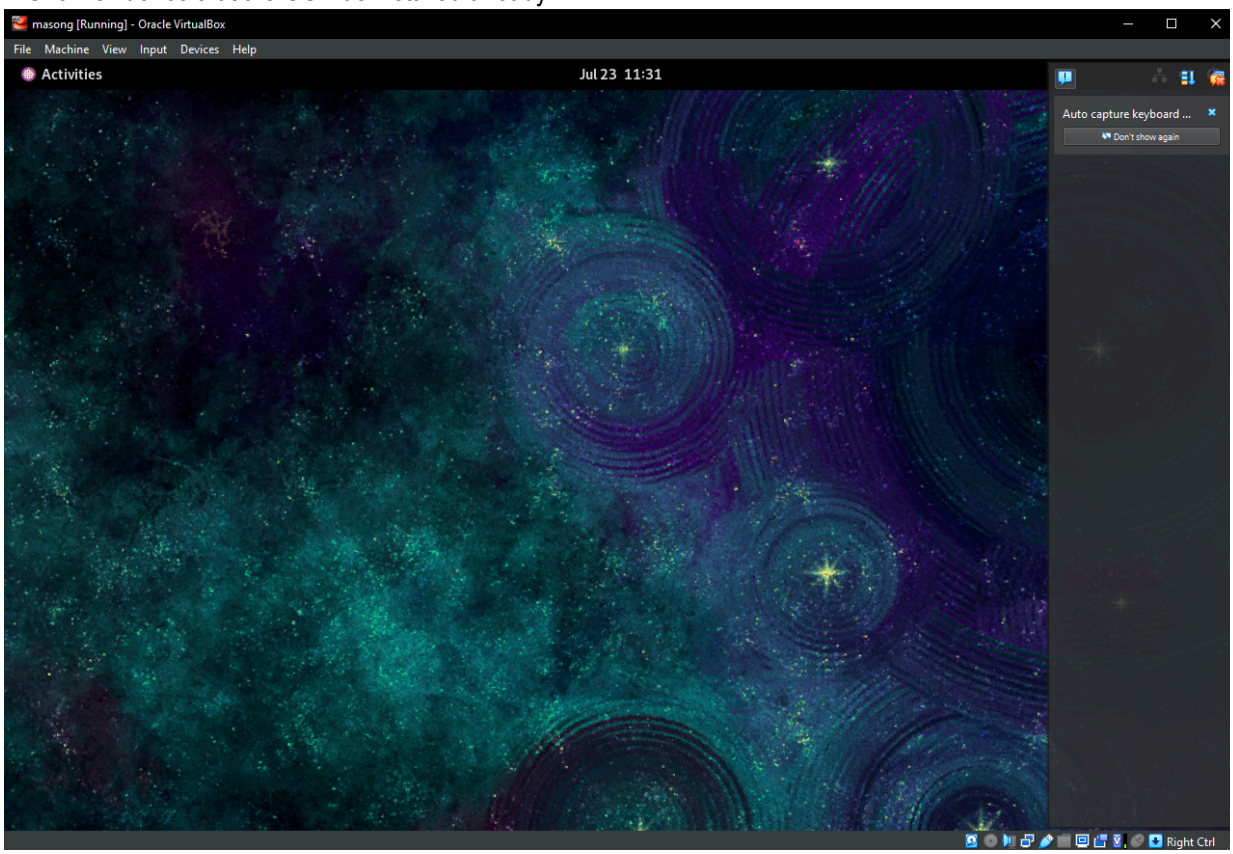




3. Install the downloaded image.

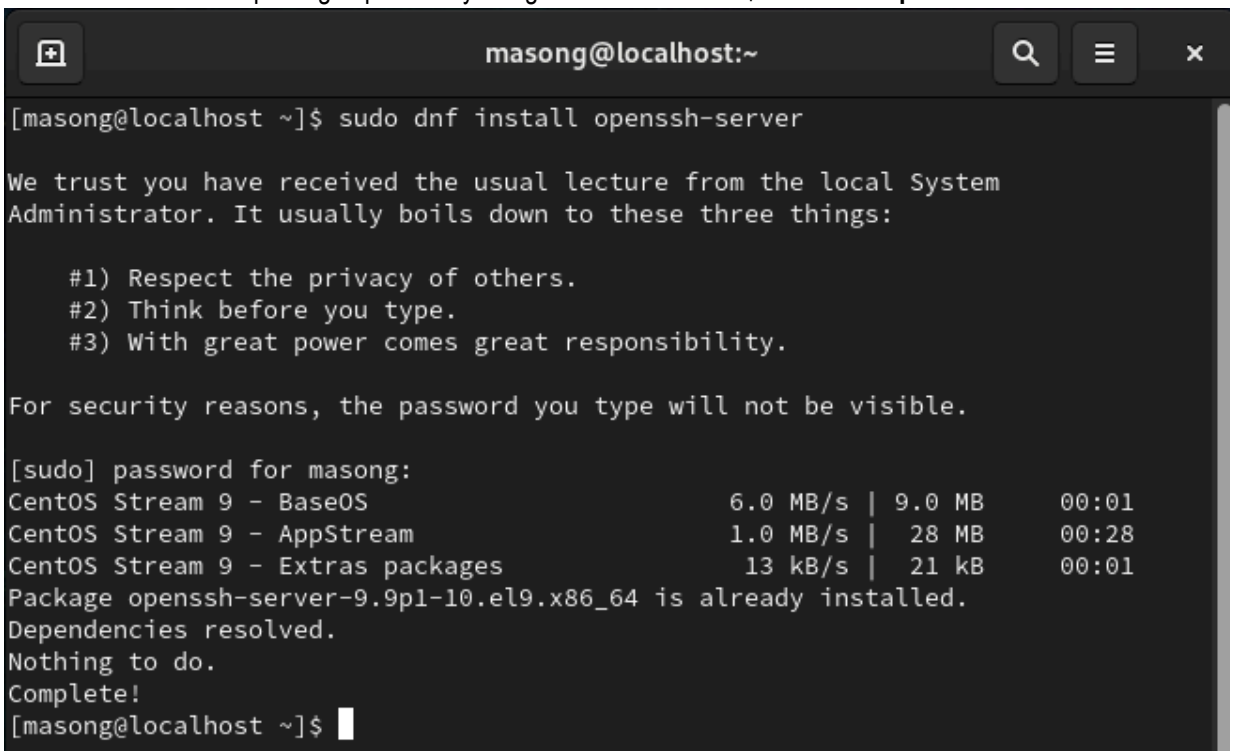


4. Show evidence that the OS was installed already.

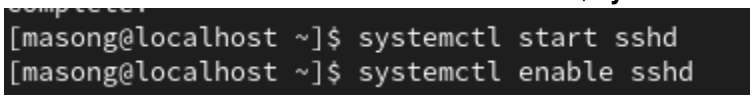


Task 2: Install the SSH server package openssh

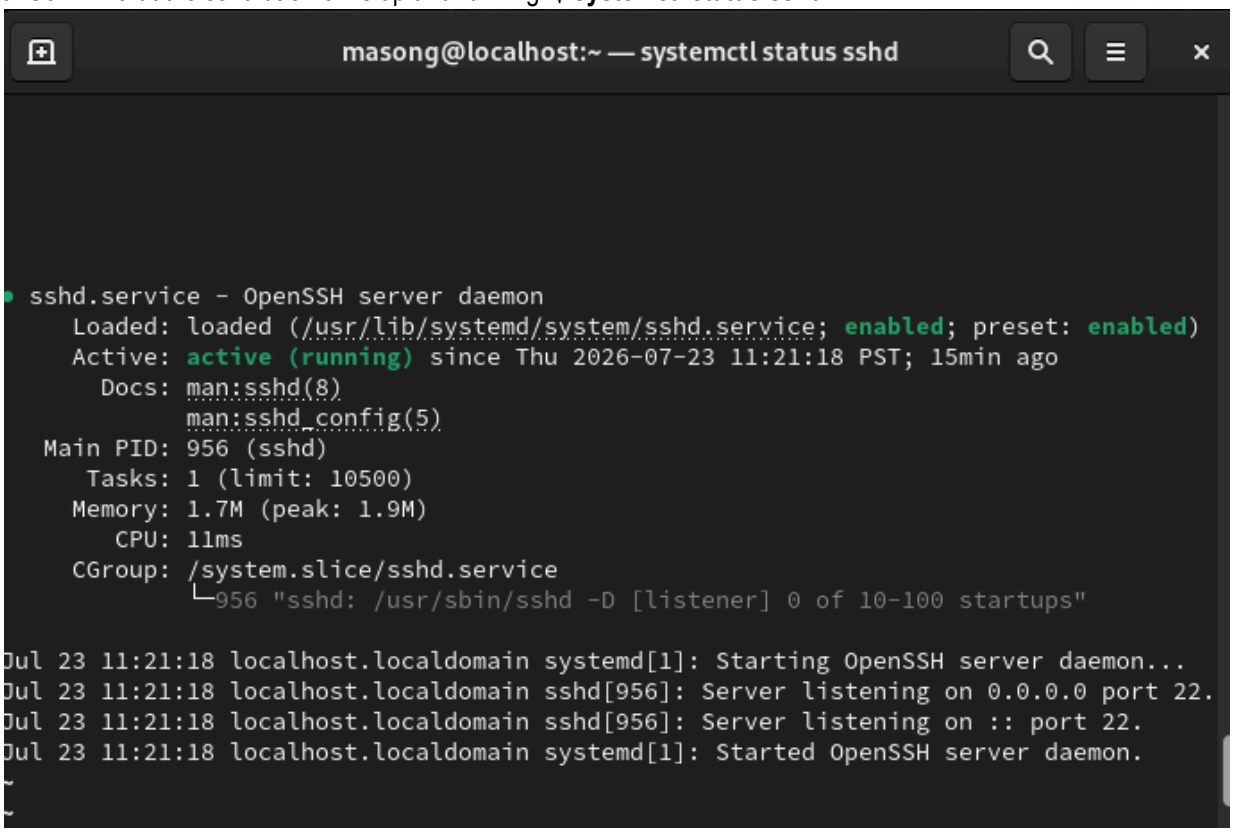
1. Install the ssh server package openssh by using the dnf command: **$ dnf install openssh-server**



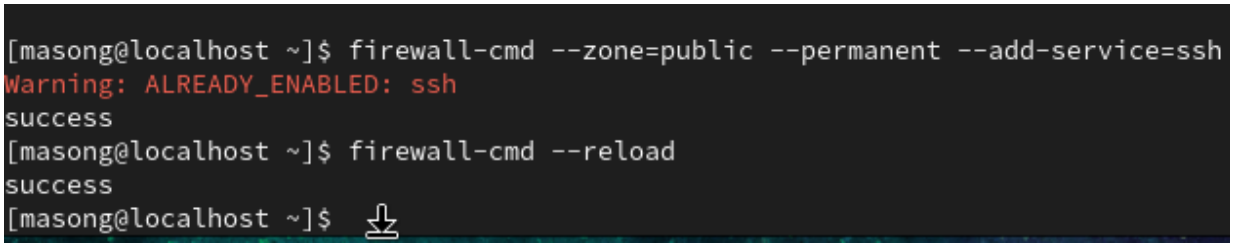
2. Start the sshd daemon and set to start after reboot: **$ systemctl start sshd, $ systemctl enable sshd**



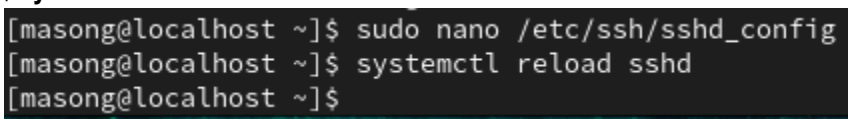
3. Confirm that the sshd daemon is up and running: **$ systemctl status sshd**



4. Open the SSH port 22 to allow incoming traffic: **$ firewall-cmd --zone=public --permanent --add-service=ssh**
$ firewall-cmd --reload



5. Locate the ssh server man config file /etc/ssh/sshd_config and perform custom configuration. Every time you make any change to the /etc/ssh/sshd-config configuration file reload the sshd service to apply changes:
$ systemctl reload sshd



Task 3: Copy the Public Key to CentOS

1. Make sure that ssh is installed on the local machine.

```
[masong@localhost ~]$ ssh -V
OpenSSH_9.9p1, OpenSSL 3.5.7 9 Jun 2026
[masong@localhost ~]$
```

2. Using the command ssh-copy-id, connect your local machine to CentOS.

```
masongsong@Masongsong:~$ ssh-copy-id masong@192.168.56.102
The authenticity of host '192.168.56.102 (192.168.56.102)' can't be established.
ED25519 key fingerprint is SHA256:cHnDD7Kd9VBCHohQrCocv0nPpud80mbZ6KAQCIKuPDI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
masong@192.168.56.102's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'masong@192.168.56.102'"
and check to make sure that only the key(s) you wanted were added.

masongsong@Masongsong:~$
```

3. On CentOS, verify that you have the authorized_keys.

```
[masong@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAACAQClPVhOGhAsxC8yRS9WjGgFY856y1r0ewhhQeU+l1X+T9r9RWXoayxa
LV/rupgBOIch+pMwrimLbekVrxhb9/zM/n1nVFje9fIB70ewKBmGmfVhayYxuEzwBPWMiT1U3XYKSiB7VzvbMaHqAI3A
UMUDrdWcVws3MtKG93Kf0lL8syCpoG8+MXkbAO8u//M4A3KkAFESHgOd+IZnwImxljGImTCwpVVNpvuxxaZ4ggOUiv/J
wK45bA0f1OatuPkFogqPPX3T66nWhVpA0IG1wGn8onB/LFhojwJN+U8sYy50twHo3LEQnBo2IsUdSoluImAfYW+nyBSW
XdURWLN+t+/uDmn8LSiPuKOQ0OQRgaRpVdSVmcmEtUESrTpP0HI5A5L2loUWsLkqqvuPFOdZ+TwLLsakwy1TCwvw62pI
uUpdGlGUm+92DnOMQj8kAYhb3KHZXTy1Js5gAVmZSsyuORY4kdOWdwVth76ByZ2d6AQhqEq14gdMAxnUmHPJDYhgkW19
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7CH5FYO6SXZq2g44nheFebn5Z+j7Twig0/e66R8Ejn/qKZQR8/xoQVQgmi8NCrtlnYGsDKcluBve/w== masongsong@
Masongsong
[masong@localhost ~]$
```

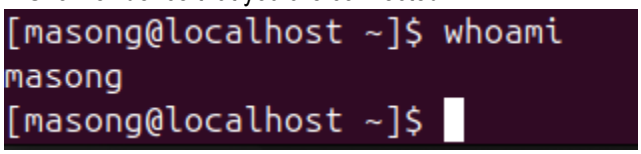
Task 4: Verify ssh remote connection

1. Using your local machine, connect to CentOS using ssh.

```
masongsong@Masongsong:~$ ssh masong@192.168.56.102
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 17:43:53 2026
[masong@localhost ~]$
```

2. Show evidence that you are connected.



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

We should be looking for distributions that we are already familiar with in order for it to function to our liking. Also both have different update times too, Debian being slow in updates but stable throughout the update, and RHEL being fast at updating but is prone to bugs in earlier versions.

2. What are the main differences between Debian and Red Hat Linux distributions?

Their package managers: for Debian (apt, apt-get, dpkg) and for RHEL (dnf, yum, rpm). Debian is for beginner to intermediate users who want to start working with the Linux Operating System while RHEL is for the more advanced versions of Linux users, this is usually used in server management, cloud server management, and DevOps.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Claude Ai**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Assistance in setting up enp0s8 in centOS because it won't appear no matter what experiment I did.**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I used Claude in order to setup my ip address in centOS since ubuntu was working just fine when I turned on host only network for both of them.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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